

Hypothesis testing with e-values

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Chapter 6: The log-optimal e-value and reverse information projection

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Recap

Simple Null vs Simple Alternative

- **Simple** null $\mathbb{P}$ vs **simple** alternative $\mathbb{Q}$ with $\mathbb{Q} \ll \mathbb{P}$
 - (Chap. 2) Admissible e-variable $\mathbb{E}^{\mathbb{P}}[E] = 1 \Rightarrow E = \frac{d\mathbb{Q}'}{d\mathbb{P}}$ likelihood ratio for some probability distribution $\mathbb{Q}' \ll \mathbb{P}$
 - (Chap. 3) Log-optimal e-variable is the likelihood ratio

$$E^* = \frac{d\mathbb{Q}}{d\mathbb{P}}$$

with e-power

$$\mathbb{E}^{\mathbb{Q}}[\log E^*] = \text{KL}(\mathbb{Q}, \mathbb{P}).$$

- E^* is the only $\mathbb{P}$ -admissible e-variable that is a likelihood ratio between $\mathbb{Q}$ and some other distribution

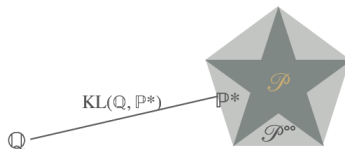
Overview of Chapter 6

- **Composite** null $\mathcal{P}$ vs **simple** alternative $\mathbb{Q}$
 - Under some assumptions, the log-optimal (numeraire) e-variable E^* is

$$E^* = \frac{d\mathbb{Q}}{d\mathbb{P}^*}$$

for some measure $\mathbb{P}^*$ equivalent to $\mathbb{Q}$, with e-power

$$\mathbb{E}^{\mathbb{Q}}[\log E^*] = \text{KL}(\mathbb{Q}, \mathbb{P}^*).$$



- E^* is the only e-variable that is a likelihood ratio between $\mathbb{Q}$ and some other measure
- E^* dominates universal inference e-variables

Notations

- **Composite** null $\mathcal{P}$ vs **simple** alternative $\mathbb{Q}$
- $\text{Conv}(\mathcal{P})$ convex hull of $\mathcal{P}$
- $\mathfrak{E}$ set of $\mathcal{P}$ -valid e-variables
- $\mathbb{Q} \ll \mathcal{P} \iff \mathbb{Q}(A) > 0 \text{ implies } \sup_{\mathbb{P} \in \mathcal{P}} \mathbb{P}(A) > 0 \text{ for every } A.$
 - We may have $\mathbb{Q} \not\ll \mathbb{P}$ for some $\mathbb{P} \in \mathcal{P}$
- Kullback-Leibler divergence

$$\text{KL}(\mathbb{Q}, \mathbb{P}) = \begin{cases} \mathbb{E}^{\mathbb{Q}}[\log \frac{d\mathbb{Q}}{d\mathbb{P}}] & \text{if } \mathbb{Q} \ll \mathbb{P} \\ \infty & \text{otherwise} \end{cases}$$

The numeraire e-variable

Definition

A **numeraire e-variable** is a $\mathbb{Q}$ -almost surely strictly positive e-variable E^* such that

$$\mathbb{E}^{\mathbb{Q}}[E/E^*] \leq 1$$

for every e-variable $E \in \mathfrak{E}$.

- **Uniqueness**: The numeraire is unique up to $\mathbb{Q}$ -nullsets.
- **Equivalence to log-optimality**

Proposition (Log-optimality of numeraire)

Let E^ be a $\mathbb{Q}$ -almost surely strictly positive e-variable. Then E^* is a numeraire if and only if it is log-optimal.*

Existence and finiteness

- **Existence.** A numeraire always exists.
- **Finiteness criterion.** The following are equivalent:
 - ① E^* is $\mathbb{Q}$ -a.s. finite.
 - ② Every $E \in \mathfrak{E}$ is $\mathbb{Q}$ -a.s. finite.
 - ③ $\mathbb{Q} \ll \mathcal{P}$.

- For any $E \in \mathfrak{E}$ and any $\mathbb{P} \in \text{Conv}(\mathcal{P})$,

$$\mathbb{E}^{\mathbb{Q}}[\log E] \leq \text{KL}(\mathbb{Q}, \mathbb{P}).$$

[LHS may be $-\inf$ and RHS $+\inf$]

- Hence

$$\sup_{E \in \mathfrak{E}} \mathbb{E}^{\mathbb{Q}}[\log E] \leq \inf_{\mathbb{P} \in \text{Conv}(\mathcal{P})} \text{KL}(\mathbb{Q}, \mathbb{P}).$$

- This motivates searching for an E and a $\mathbb{P}$ that jointly achieve equality.

Strong duality for finite nulls

- Suppose $\mathcal{P} = \{\mathbb{P}_1, \dots, \mathbb{P}_K\}$ with each $\mathbb{P}_k \ll \mathbb{Q}$ and $\min_{\mathbb{P} \in \text{Conv}(\mathcal{P})} \text{KL}(\mathbb{Q}, \mathbb{P}) < \infty$.
- Then

$$\mathbb{E}^{\mathbb{Q}}[\log E^*] = \max_{E \in \mathfrak{E}} \mathbb{E}^{\mathbb{Q}}[\log E] = \min_{\mathbb{P} \in \text{Conv}(\mathcal{P})} \text{KL}(\mathbb{Q}, \mathbb{P}) = \text{KL}(\mathbb{Q}, \mathbb{P}^*),$$

where $\mathbb{P}^* = \sum_{k=1}^K \lambda_k^* \mathbb{P}_k$ is the **Reverse information projection** (RIPr) of $\mathbb{Q}$ onto $\text{Conv}(\mathcal{P})$ and the numeraire equals the likelihood ratio

$$E^* = \frac{d\mathbb{Q}}{d\mathbb{P}^*}.$$

Strong duality for finite nulls

Example

- Let $\mathcal{P} = \{\mathcal{N}(-1, 1), \mathcal{N}(1, 1)\}$ and $\mathbb{Q} = \mathcal{N}(0, 1)$
- $\text{Conv}(\mathcal{P}) = \{\lambda\mathcal{N}(-1, 1) + (1 - \lambda)\mathcal{N}(1, 1) | \lambda \in [0, 1]\}$
- RIPr

$$\mathbb{P}^* = \frac{1}{2}\mathcal{N}(-1, 1) + \frac{1}{2}\mathcal{N}(1, 1)$$

- Numeraire

$$E^*(Z) = \frac{\phi(Z)}{\frac{1}{2}(\phi(Z+1) + \phi(Z-1))} = \frac{e^{1/2}}{\cosh(Z)}.$$

Strong duality for finite nulls

Example 2

- Let $\mathcal{P} = \{\mathcal{N}(-1, 1), \mathcal{N}(0, 1)\}$ and $\mathcal{Q} = \mathcal{N}(1, 1)$
- $\text{Conv}(\mathcal{P}) = \{\lambda\mathcal{N}(-1, 1) + (1 - \lambda)\mathcal{N}(0, 1) | \lambda \in [0, 1]\}$
- RIPr

$$\mathbb{P}^* = \mathcal{N}(0, 1) \in \mathcal{P}$$

- Numeraire

$$E^*(Z) = \frac{\phi(Z - 1)}{\phi(Z)}.$$

Strong duality for finite nulls

Numeraire is a Bayes factor

- $H_0 : Z \sim \mathbb{P}_c$ with $c \sim \text{Discrete}(\lambda_1^*, \dots, \lambda_K^*)$ vs $H_1 : Z \sim \mathbb{Q}$

$$E^* = \frac{d\mathbb{Q}}{\sum_{k=1}^K \lambda_k^* \mathbb{P}_k}.$$

Reverse information projection (RIPr)

Definition (RIPr (general definition, no finiteness/DOM assumptions))

Define the **RIPr** $\mathbb{P}^* \ll \mathbb{Q}$ of $\mathbb{Q}$ onto $\mathcal{P}$ by

$$\frac{d\mathbb{P}^*}{d\mathbb{Q}} = \frac{1}{E^*}, \quad \text{with } E^* \text{ the numeraire.}$$

- $\mathbb{P}^*$ is a (sub-)probability measure lying in the **bipolar** $\mathcal{P}^{\circ\circ}$ of $\mathcal{P}$ (the *effective null*)

$$\mathcal{P}^{\circ\circ} = \{\mathbb{P} \in \mathcal{M}_+ : \mathbb{E}^{\mathbb{P}}[E] \leq 1 \ \forall E \in \mathfrak{E}\}.$$

- Interpretation: $\mathcal{P}^{\circ\circ}$ is the largest family of sub-probability measures against which no e-variable in $\mathfrak{E}$ provides evidence; thus the **effective null**.
- Powered existence: A powered e-variable (or test) for $\mathcal{P}$ against $\mathbb{Q}$ exists
 $\iff \mathbb{Q} \notin \mathcal{P}^{\circ\circ}.$

Bipolar: examples

- $\mathcal{P} = \{\mathcal{N}(\mu, 1), \mu < 0\}$
 - $\mathcal{N}(0, 1) \notin \text{Conv}(\mathcal{P})$ but $\mathcal{N}(0, 1) \in \mathcal{P}^{\circ\circ}$
- $\mathcal{P} = \{\delta_x : x \in [0, 0.5]\}$
 - $\mathcal{P}^{\circ\circ}$ contains all bounded distributions on $[0, 0.5]$

Strong duality (general, dominated case)

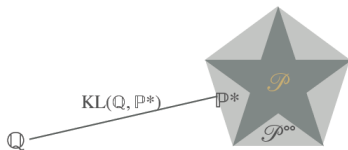
- General strong duality. If $\mathbb{Q} \ll \mathcal{P}$,

$$\sup_{E \in \mathfrak{E}} \mathbb{E}^{\mathbb{Q}}[\log E] = \inf_{\mathbb{P} \in \mathcal{P}^{\circ\circ}} \text{KL}(\mathbb{Q}, \mathbb{P}) = \text{KL}(\mathbb{Q}, \mathbb{P}^*)$$

where $\mathbb{Q} \ll \mathbb{P}^*$ and

$$E^* = \frac{d\mathbb{Q}}{d\mathbb{P}^*}.$$

- $\mathbb{P}^*$ is the closest null-like distribution to $\mathbb{Q}$ in KL



Numerator as a likelihood ratio (and uniqueness)

- Verification theorem.
- Assume $\mathbb{Q} \ll \mathcal{P}$. Let $E^* > 0$ and define $d\mathbb{P}^*/d\mathbb{Q} = 1/E^*$.
- Then

$$E^* \text{ is numerator} \iff \mathbb{P}^* \in \mathcal{P}^{\circ\circ}.$$

- E^* is the **only** e-variable that is a likelihood ratio $d\mathbb{Q}/d\mathbb{P}$ for some $\mathbb{P} \in \mathcal{P}^{\circ\circ}$ equivalent to $\mathbb{Q}$.

How to identify a numeraire in practice

- 1 Guess a candidate likelihood ratio dQ/dP against an *extreme* P in $\mathcal{P}^{\circ\circ}$.
- 2 Verify it is an e-variable for $\mathcal{P}$.

Numerator dominates universal inference

- Assume a common reference measure L ; write densities in lowercase.
- For point alternative q and null $\mathcal{P}$,

$$E_{\text{UI}}(Z) = \frac{q(Z)}{p_{\max}(Z)}, \quad p_{\max}(z) = \sup_{p \in \mathcal{P}} p(z),$$

- Numeraire is

$$E^*(Z) = \frac{q(Z)}{p^*(Z)}$$

with p^* the RPr density.

- One can show $p^*(Z) \leq p_{\max}(Z)$ (a.s.), hence $E^*(Z) \geq E_{\text{UI}}(Z)$, often strictly.
- Universal-inference e-variable is inadmissible, and the numeraire yields (weakly) larger e-power with weaker assumptions.

Numerator dominates universal inference

Example

- Let $\mathcal{P} = \{\mathcal{N}(-1, 1), \mathcal{N}(1, 1)\}$ and $\mathbb{Q} = \mathcal{N}(0, 1)$
- $\text{Conv}(\mathcal{P}) = \{\lambda\mathcal{N}(-1, 1) + (1 - \lambda)\mathcal{N}(1, 1) | \lambda \in [0, 1]\}$
- Numeraire

$$E^*(Z) = \frac{\phi(Z)}{\frac{1}{2}(\phi(Z+1) + \phi(Z-1))} = \frac{e^{1/2}}{\cosh(Z)}.$$

- Universal Inference

$$E_{\text{UI}}(Z) = \frac{\phi(Z)}{\phi(Z - \text{sign}(Z))}$$

- We have $p_{\max}(z) = \phi(z - \text{sign}(z)) > \frac{1}{2}(\phi(z+1) + \phi(z-1)) = p^*(z)$ for all $z \neq 0$ hence $E_{\text{UI}}(Z) < E^*(Z)$ $\mathcal{P}$ -a.s.

E-values and Bayesian inference

- $X_1, \dots, X_n$ i.i.d. from some unknown distribution p_0
- Composite null $\mathcal{P}$ vs composite alternative $\mathcal{Q}$
- Frequentist: $H_0 : p_0 \in \mathcal{P}$ vs $H_1 : p_0 \in \mathcal{Q}$
- (Full) Bayesian: $H_0 : p_0 = p, p \sim \nu$ vs $H_1 : p_0 = q, q \sim \mu$

Bayes factor, numeraire, generalised likelihood ratio

- Bayes factor

$$B_n = \frac{\int_{\mathcal{Q}} \prod_{i=1}^n q(X_i) \mu(dq)}{\int_{\mathcal{P}} \prod_{i=1}^n p(X_i) \nu(dp)},$$

- Generalised likelihood ratio

$$G_n = \frac{\sup_{q \in \mathcal{Q}} \prod_{i=1}^n q(X_i)}{\sup_{p \in \mathcal{P}} \prod_{i=1}^n p(X_i)}.$$

- Universal Inference (method of mixtures)

$$U_n = \frac{\int_{\mathcal{Q}} \prod_{i=1}^n q(X_i) \mu(dq)}{\sup_{p \in \mathcal{P}} \prod_{i=1}^n p(X_i)},$$

- Numeraire

$$E_n = \frac{\int_{\mathcal{Q}} \prod_{i=1}^n q(X_i) \mu(dq)}{p^*(X_1, \dots, X_n)}.$$

Example: One-parameter exponential family

- Let $p_\theta(z) = \exp\{\theta T(z) - A(\theta)\}$ with null $\Theta_0 = (-\infty, \theta_0]$ and point alternative $\theta_1 > \theta_0$.
- The RPr is p_{θ_0} and the numeraire is the likelihood ratio

$$E^*(Z) = \frac{p_{\theta_1}(Z)}{p_{\theta_0}(Z)} = \exp\{(\theta_1 - \theta_0)T(Z) - A(\theta_1) + A(\theta_0)\}.$$

Example: Testing symmetry

- For $Z \in \mathbb{R}$, null: $Z \stackrel{d}{=} -Z$.
- For an alternative with density q , the RPr is the symmetrization

$$p^*(z) = \frac{1}{2}\{q(z) + q(-z)\}$$

- Then

$$E^*(Z) = \frac{q(Z)}{p^*(Z)} = \frac{2q(Z)}{q(Z) + q(-Z)}.$$

Example: Testing bounded means on $[0, 1]$

- Null: $\mathbb{E}^{\mathbb{P}}[Z] \leq \mu$ for $Z \in [0, 1]$; fix alternative $\mathbb{Q}$ with density q (e.g., uniform for illustration).
- Look for the log-optimal e-variable within the class $1 + \lambda(Z - \mu)$:

$$E^*(Z) = 1 + \lambda^*(Z - \mu), \quad \lambda^* > 0,$$

where λ^* solves the first-order condition

$$\mathbb{E}^{\mathbb{Q}} \left[\frac{Z - \mu}{1 + \lambda^*(Z - \mu)} \right] = 0.$$

- Equivalently, $\frac{1 + \lambda^*(1 - \mu)}{1 - \lambda^*\mu} = e^{\lambda^*}$ for $\mathbb{Q} = \text{Unif}[0, 1]$.
- This E^* is strictly positive and is the numeraire by Corollary 6.16.

Takeaways

- For composite null $\mathcal{P}$ vs simple alternative $\mathbb{Q}$
- A **log-optimal** e-variable (the numeraire) exists and is unique up to $\mathbb{Q}$ -nullsets.
- The numeraire pairs with the **RIPr** to yield **strong duality**: maximize $\mathbb{E}^{\mathbb{Q}}[\log E]$
 $\leftrightarrow$ minimize $\text{KL}(\mathbb{Q}, \cdot)$ over the effective null.
- The numeraire is the only e-variable that is also a likelihood ratio $d\mathbb{Q}/d\mathbb{P}$ for some $\mathbb{P}$.
- It **dominates universal inference** (pointwise and in e-power) under mild conditions.
- Concrete, interpretable forms arise in exponential families, monotone LR models, symmetry, bounded means, and more.